Dr B R Ambedkar National Institute of Technology Jalandhar Punjab

Course Title: Computer Graphics and Image Processing Laboratory

Course Code: CSDC0332

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Experiment

2D Transformations and Composite Transformations

Task

- Write a program to perform **2D geometric transformations** on a polygon and display the transformed object.
- Your program must perform the following operations:
  - **Translation**
  - **Scaling**
  - **Rotation**
  - **Rotation about an Arbitrary Point**
  - **Scaling about an Arbitrary Point**
  - **Composite Transformation (Scaling → Rotation → Translation)**
- Display the **original polygon and transformed polygon**.

Input Polygon

Use the following **hexagonal irregular polygon**:

A (2,2)
B (5,1)
C (8,3)
D (7,6)
E (4,8)
F (1,5)

Transformation Parameters

Translation

- tx = 3
  ty = 2

Scaling

- Sx = 2
  Sy = 1.5

Rotation

- $\theta = 45°$

Arbitrary Point

- P (4,4)

Tasks to Perform

- **Task 1: Translation**
  - Translate the polygon using

    tx = 3
    ty = 2
  - Display:
  - Original polygon
  - Translated polygon

Task 2: Scaling about Origin

- Scale the polygon using
- Sx = 2
  Sy = 1.5
- Display:
- Original polygon
- Scaled polygon

Task 3: Rotation about Origin

- Rotate the polygon using
- $\theta = 45°$
- Display:
- Original polygon
- Rotated polygon

Task 4: Rotation about Arbitrary Point

- Rotate the polygon about point
  - P (4,4)
- Perform the following sequence:
  1. Translate polygon by (-4,-4)
  2. Rotate by 45°
  3. Translate polygon by (4,4)
- Display:
- Original polygon
- Rotated polygon about arbitrary point

Task 5: Scaling about Arbitrary Point

- Scale the polygon about point
  - P (4,4)
- Perform the following sequence:
  1. Translate polygon by (-4,-4)
  2. Apply scaling (Sx = 2, Sy = 1.5)
  3. Translate polygon by (4,4)
- Display:
- Original polygon
- Scaled polygon about arbitrary point

Task 6: Composite Transformation

- Perform the following transformations in sequence:
  1. Scaling (Sx = 2, Sy = 1.5)
  2. Rotation (45°)
  3. Translation (tx = 3, ty = 2)
- Display:
- Original polygon
- Final transformed polygon

Program Requirements

Your program must:

accept polygon coordinates,

apply transformation matrices,

display both **original and transformed polygons**,

use **different colors for each transformation**,

label the vertices before and after transformation.

Expected Output

The program should display:

Original polygon

Translated polygon

Scaled polygon

Rotated polygon

Rotation about arbitrary point

Scaling about arbitrary point

Composite transformation result